

Birla Institute of Technology & Science, Pilani
First Semester 2023-2024 (Comprehensive Examination)
ECE F314: Electromagnetic Field & Microwave Engineering
Part-A (Closed Book)

Duration: 90 minutes

Total Marks: 45

Note to Students:

Please write the answers in the answer sheet provided with proper units and signs. Marks will be deducted for improper signs and units.

1.
 - a) In an impedance **Smith chart**, a clockwise movement along a constant resistance circle gives rise to (decrease/increase) in the value of reactance. [1]
 - b) The basic mechanism of the **Klystron amplifier** is based on the principle of modulation. [1]
 - c) The function of **ferrite** material in a circulator is [1]
 - d) If the phase difference between voltage and current in a device is **180°**, such a device we studied in this course is [1]
 - e) In a transmission line, the reflection coefficient domain is a circle of unitary radius. [1]
 - f) In a good conductor, leads by **45°**. [1]
 - g) In the TEM mode, **E_z** =, and **H_z** = [1]
 - h) To get **100%** efficiency from an antenna, the values of antenna directivity and antenna gain are..... [1]
 - i) The Smith chart represents a family of **arcs** obtained by varying normalized reactance in the range of [1]
 - j) A **10 km** long transmission line has a capacitance of **56 mF**. Its distributed capacitance is [1]
2.
 - a) For vacuum based high power amplifiers, the RF signal is stationary in while the RF signal travels along the length in [2]
 - b) What happens to the reflected and transmitted light when an unpolarized wave hits the interface at **Brewster's angle**? [2]
 - c) Write down the S matrix for an **ideal and reciprocal attenuator**. [2]
 - d) A power of **1 W** is sent from an input terminal of a transmission line. The power loss measured at the output terminal of the transmission line is **-3 dB**. If the input voltage was '**Y**' volts, what would be the voltage at the output? [2]
 - e) What types of amplifiers use the **slow wave structures**, and what is their use in amplification? [2]
 - f) The frequency generated in an **8-cavity magnetron** is **9.52 x 10⁹ Hz**. How much magnetic flux is required to generate this frequency? [2]
 - g) Draw the field pattern of **TE₂₃ mode** (Front view-XY plane). [2]
 - h) The operating frequency of a two-cavity klystron is **10 GHz**. The current and voltage applied to the Klystron are **14.4 mA** and **40kV**, respectively. Find the transit angle in drift space if the drift space length is **1 cm**. [2]

3. a) A parallel-plate waveguide will be used to propagate an EM wave of frequency **4GHz**. The waveguide is filled with a lossless dielectric ($\epsilon_r = 2.25$ & $\mu_r = 1$), and the plate separation is **4 cm**. Find the propagating modes in the waveguide. [3]
- b) A sinusoidal electric intensity of amplitude **100 V/m** and frequency **2 GHz** exists in a lossy dielectric medium with a relative permittivity of **2.5** and a loss tangent of **0.005**. Find the average power dissipated in the medium per cubic meter. [3]
- c) A **5 MHz** plane wave traveling in a normally dispersive lossless medium has a phase velocity of **250×10^6 m/s**. Phase velocity is a function of λ , given as $K \cdot \lambda^{1/2}$, where K is a constant. Find the group velocity. [3]
- d) Write the expressions for electric and magnetic field intensities of a sinusoidally time-varying uniform plane wave propagating in free space having the following characteristics: **(a) $f=100\text{MHz}$** **(b) direction of propagation in $+z$ direction** **(c) polarization is right circular with electric field in $z=0$ plane at $t=0$ having x component = E_0 and y component = $0.75 E_0$.** [3]
- e) For a **12-cavity** magnetron, draw the electric lines pattern for $\pi/2$ mode operation and mention the **number of spokes** generated during this operation. [3]
4. For the TE_{11} mode, Given that $f = 10 \text{ GHz}$, $a=b = 2.236 \text{ cm}$, $H_z = H_0 \cos((\pi/a) \cdot x) \cos((\pi/b) \cdot y)$ A/m and $E_z = 0$. Find the value of f_c , V_p , λ_g , and write the expression for H_y . [4]

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1. Consider a **5mm** thick glass slab ($\epsilon_r = 2.56$); a beam of green light (free space wavelength = **532 nm**) is incident on the glass normally.
 - a) What color would it appear to a sensor embedded in glass? What is the frequency of the corresponding EM wave inside the glass?
 - b) What percentage of incident power is reflected by and transmitted into the glass? **[4]**
2. In **E** plane Tee, assuming the system is reciprocal and port 3 is perfectly matched. Write down the values of **S** matrix elements for E plane Tee using the lossless property of the network. **[4]**
3. The guided wavelength for a frequency of **20 GHz** is **6 cm** when the dominant mode is propagating in an air-filled waveguide. Find the dimension of the rectangular waveguide assuming **a=2.5b**. **[5]**
4. The electric and magnetic fields in a coaxial cable with its axis along the z and with inner and outer radii a and b, respectively, are given by $\mathbf{E} = (V/(r \ln(b/a))) * \cos(\omega t - z(\mu_0 \epsilon_0)^{1/2}) \mathbf{a}_r$ and $\mathbf{H} = (I/(2\pi r)) * \cos(\omega t - z(\mu_0 \epsilon_0)^{1/2}) \mathbf{a}_\phi$ for $a < r < b$. **V** and **I** are constants. Find the time average power flow along the coaxial cable. **[5]**
5. A directional coupler has four ports that are perfectly matched; the network is symmetric, ports **1 and 3**, and ports **2 and 4** are isolated. Using the lossless property, find out the values of the S matrix of the directional coupler. **[5]**
6. The direction of the Magnetic Field vector of a plane wave is represented by $\mathbf{H} = 5\mathbf{a}_y - 8\mathbf{a}_z$. The phase constant of the wave at **1GHz** is **25 rad/m**; find the complete electric field of the wave. Comment on its polarization. **[6]**
7. Radio Mirchi broadcast (**98.3 MHz**) station uses a **300 Ω** cable between a matched transmitter and a tower-mounted half-wave dipole antenna. The antenna impedance is **73 Ω**. The transmitter is designed for an output power of **0.1 kW**. What is the power delivered to the antenna? The designer chooses to use a quarter-wave section to match the impedance between the transmitter and the antenna. For the quarter-wave section, what should be the length and characteristic impedance? **[6]**
8. The electric field associated with a plane wave is given by $\mathbf{E} = (7.83 \mathbf{a}_x - 4.5 \mathbf{a}_z) e^{-j7(0.5\mathbf{a}_x + 0.87\mathbf{a}_z)}$. If this field is incident on a perfectly conducting plane oriented perpendicular to the z-axis, find the total electric and magnetic field in front of the perfect conductor. **[10]**